

Module 3 Activity 3: Using the Insert Cursor

Purpose:

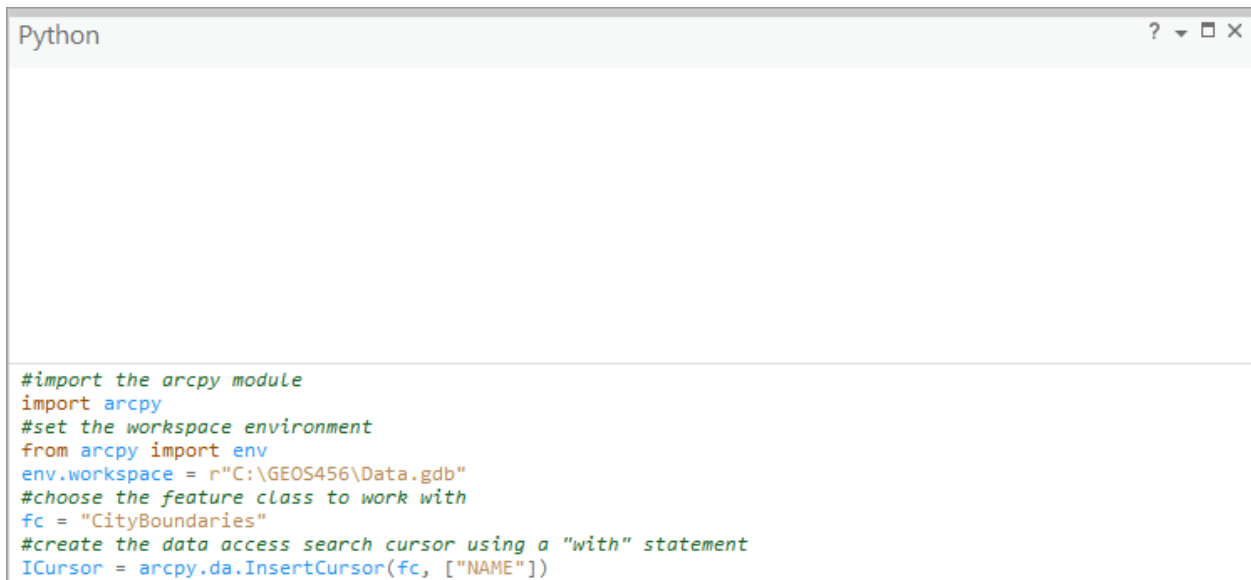
Use the ***InsertCursor()*** function to insert rows in a table and to populate those rows with information

Instructions:

You can use any Python editor for this exercise.

Part 1:

1. Import the ***arcpy*** module
2. Set the workspace
3. Choose which feature class you want to work with
4. Create the ***InsertCursor*** inside a variable called ***ICursor***
5. Reference the feature class in the ***InsertCursor*** syntax and specify the field you want to search. Fields are a required parameter in the ***InsertCursor*** and are referenced using square brackets ***[]***



```
Python
#import the arcpy module
import arcpy
#set the workspace environment
from arcpy import env
env.workspace = r"C:\GEOS456\Data.gdb"
#choose the feature class to work with
fc = "CityBoundaries"
#create the data access search cursor using a "with" statement
ICursor = arcpy.da.InsertCursor(fc, ["NAME"])
```

In order to insert multiple new rows, you must define a variable to tell the cursor how many rows to insert.

6. Create a variable ***x*** and define the count number as ***1***

```
Python
#import the arcpy module
import arcpy
#set the workspace environment
from arcpy import env
env.workspace = r"C:\GEOS456\Data.gdb"
#specify the feature class to work with
fc = "CityBoundaries"
#create the data access search cursor using a "with" statement
ICursor = arcpy.da.InsertCursor(fc, ["NAME"])
#define the count number interval with a variable
x = 1
```

7. Start a **while** loop specifying the number of rows to be inserted
8. Use the **insertRow()** method to reference the **InsertCursor** and populate the new row with user entered values

NOTE: the values entered in the new row must match the data type of the existing field

9. Define the intervals of the rows inserted, in this case **1**

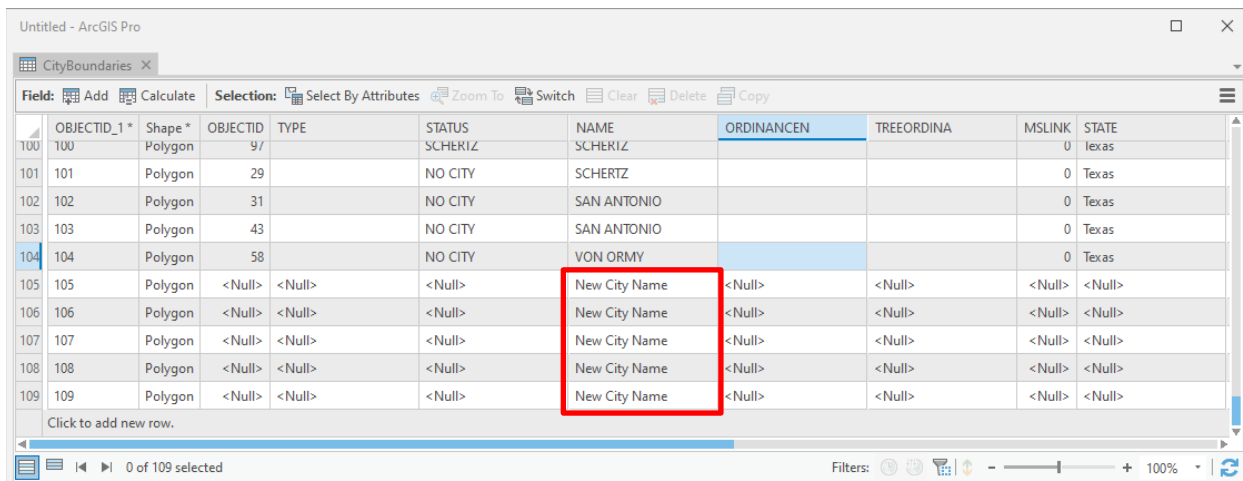
```
Python
#import the arcpy module
import arcpy
#set the workspace environment
from arcpy import env
env.workspace = r"C:\GEOS456\Data.gdb"
#specify the feature class to work with
fc = "CityBoundaries"
#create the data access search cursor using a "with" statement
ICursor = arcpy.da.InsertCursor(fc, ["NAME"])
#define the count number interval with a variable
x = 1
#start a WHILE loop to determine how many rows will be inserted
while x <= 5:
    #use the "insertRow" method to populate the NAME field
    ICursor.insertRow(["New City Name"])
    #the following adds one row at a time
    x += 1
```

10. Use the **del** function to delete the cursor object and remove any locks on the data

```
Python
#import the arcpy module
import arcpy
#set the workspace environment
from arcpy import env
env.workspace = r"C:\GEOS456\Data.gdb"
#specify the feature class to work with
fc = "CityBoundaries"
#create the data access search cursor using a "with" statement
Icursor = arcpy.da.InsertCursor(fc, ["NAME"])
#define the count number interval with a variable
x = 1
#start a WHILE loop to determine how many rows will be inserted
while x <= 5:
    #use the "insertRow" method to populate the NAME field
    Icursor.insertRow(["New City Name"])
    #the following adds one row at a time
    x += 1
#delete the cursor object to remove locks on the data
del Icursor
```

11. Run the script
12. Open ArcMap or ArcCatalog and examine your results

Upon examining your results, you will see that **5 new rows** have been inserted at the bottom of the existing rows. As you only specified a value for the **NAME** field, only those will have values. The remaining fields will also have rows inserted, but those rows will be a value of **<null>**



OBJECTID_1*	Shape *	OBJECTID	TYPE	STATUS	NAME	ORDINANCEN	TREEORDINA	MSLINK	STATE
100	Polygon	97		SCHERIZ	SCHERIZ			0	Texas
101	Polygon	29		NO CITY	SCHERTZ			0	Texas
102	Polygon	31		NO CITY	SAN ANTONIO			0	Texas
103	Polygon	43		NO CITY	SAN ANTONIO			0	Texas
104	Polygon	58		NO CITY	VON ORMY			0	Texas
105	Polygon	<Null>	<Null>	<Null>	New City Name	<Null>	<Null>	<Null>	<Null>
106	Polygon	<Null>	<Null>	<Null>	New City Name	<Null>	<Null>	<Null>	<Null>
107	Polygon	<Null>	<Null>	<Null>	New City Name	<Null>	<Null>	<Null>	<Null>
108	Polygon	<Null>	<Null>	<Null>	New City Name	<Null>	<Null>	<Null>	<Null>
109	Polygon	<Null>	<Null>	<Null>	New City Name	<Null>	<Null>	<Null>	<Null>

The screenshot above shows the additional rows have been added and only the NAME field has new values populated.

To add a unique value to each new row, you can add an **if** statement inside the **while** loop. When doing so, you must specify which row is to be populated with each unique name using a conditional statement. See below.

```
Python
#import the arcpy module
import arcpy
#set the workspace environment
from arcpy import env
env.workspace = r"C:\GEOS456\Data.gdb"
#specify the feature class to work with
fc = "CityBoundaries"
#create the data access search cursor using a "with" statement
ICursor = arcpy.da.InsertCursor(fc, ["NAME"])
#define the count number interval with a variable
x = 1
#start a while loop to determine how many rows to be inserted
while x <=5:
    if x == 1:
        ICursor.insertRow(["Calgary"])
    elif x == 2:
        ICursor.insertRow(["Edmonton"])
    elif x == 3:
        ICursor.insertRow(["Red Deer"])
    elif x == 4:
        ICursor.insertRow(["Lethbridge"])
    elif x == 5:
        ICursor.insertRow(["Banff"])
    #the following adds one row at a time
    x += 1
#delete the cursor object to remove locks on the data
del ICursor
```

Once the script is executed with the *if* statement, it will insert the rows with the unique values.

Note: If you just re-run the script, you will notice that the inserted rows appear below the ones already inserted using the steps above. This is because cursors only move forward and not backwards.

Untitled - ArcGIS Pro

CityBoundaries

Field: Add Calculate Selection: Select By Attributes Zoom To Switch Clear Delete Copy

	OBJECTID_1*	Shape *	OBJECTID	TYPE	STATUS	NAME	ORDINANCEN	TREEORDINA	MSLINK	STATE
105	105	Polygon	<Null>	<Null>	<Null>	New City Name	<Null>	<Null>	<Null>	<Null>
106	106	Polygon	<Null>	<Null>	<Null>	New City Name	<Null>	<Null>	<Null>	<Null>
107	107	Polygon	<Null>	<Null>	<Null>	New City Name	<Null>	<Null>	<Null>	<Null>
108	108	Polygon	<Null>	<Null>	<Null>	New City Name	<Null>	<Null>	<Null>	<Null>
109	109	Polygon	<Null>	<Null>	<Null>	New City Name	<Null>	<Null>	<Null>	<Null>
110	110	Polygon	<Null>	<Null>	<Null>	Calgary	<Null>	<Null>	<Null>	<Null>
111	111	Polygon	<Null>	<Null>	<Null>	Edmonton	<Null>	<Null>	<Null>	<Null>
112	112	Polygon	<Null>	<Null>	<Null>	Red Deer	<Null>	<Null>	<Null>	<Null>
113	113	Polygon	<Null>	<Null>	<Null>	Lethbridge	<Null>	<Null>	<Null>	<Null>
114	114	Polygon	<Null>	<Null>	<Null>	Banff	<Null>	<Null>	<Null>	<Null>

Click to add new row.

0 of 114 selected

Filters: 100%

Next Steps:

Modify your script created in this exercise to populate the **STATUS** field with a **NO CITY** value and the **STATE** field with a value of **Texas**.

Untitled - ArcGIS Pro

CityBoundaries

Field: Add Calculate Selection: Select By Attributes Zoom To Switch Clear Delete Copy

	OBJECTID_1 *	Shape *	OBJECTID	TYPE	STATUS	NAME	ORDINANCEN	TREEORDINA	MSLINK	STATE
110	110	Polygon	<Null>	<Null>	<Null>	Calgary	<Null>	<Null>	<Null>	<Null>
111	111	Polygon	<Null>	<Null>	<Null>	Edmonton	<Null>	<Null>	<Null>	<Null>
112	112	Polygon	<Null>	<Null>	<Null>	Red Deer	<Null>	<Null>	<Null>	<Null>
113	113	Polygon	<Null>	<Null>	<Null>	Lethbridge	<Null>	<Null>	<Null>	<Null>
114	114	Polygon	<Null>	<Null>	<Null>	Banff	<Null>	<Null>	<Null>	<Null>
115	115	Polygon	<Null>	<Null>	NO CITY	New_Name_1	<Null>	<Null>	<Null>	Texas
116	116	Polygon	<Null>	<Null>	NO CITY	New_Name_2	<Null>	<Null>	<Null>	Texas
117	117	Polygon	<Null>	<Null>	NO CITY	New_Name_3	<Null>	<Null>	<Null>	Texas
118	118	Polygon	<Null>	<Null>	NO CITY	New_Name_4	<Null>	<Null>	<Null>	Texas
119	119	Polygon	<Null>	<Null>	NO CITY	New_Name_5	<Null>	<Null>	<Null>	Texas

Click to add new row.

0 of 119 selected

Filters: 100%